

Deep learning in defects detection of PV modules: A review

Katleho Masita^{a,*}, Ali Hasan^a, Thokozani Shongwe^a, Hasan Abu Hilal^b

^a Electrical and Electronic Engineering Science Department, University of Johannesburg, Johannesburg, South Africa

^b Electrical and Electronic Engineering Department, Eastern Mediterranean University, North Cyprus, 99628, Famagusta, Cyprus

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ABSTRACT

Identifying defects in photovoltaic (PV) modules is essential for ensuring optimal performance and prolonging their operational lifespan. Traditional manual inspection methods are time-consuming, labor-intensive, and subject to human error, necessitating the development of automated, efficient detection techniques. With the increasing scale of PV power plants, there is a pressing need for automated, accurate, and efficient defect detection methods. This review explores the application of deep learning (DL) methods, particularly convolutional neural networks (CNNs), in the identification and classification of PV module defects. Various imaging techniques, including electroluminescence (EL), thermal, and visible spectrum imaging, are discussed for their roles in data acquisition. The importance of preprocessing steps such as image normalization, registration, and segmentation is emphasized to enhance detection accuracy. The review highlights the effectiveness of DL models like MobileNet, VGG-16, and YOLO, and techniques such as transfer learning and data augmentation in improving model performance. Despite achieving high accuracy, challenges such as the need for large datasets and model generalization across different PV modules and environmental conditions remain. The integration of DL with aerial inspection technologies and advancements in image processing holds promise for further enhancing the reliability and efficiency of solar energy systems.

1. Introduction

The Deep learning (DL) has emerged as a powerful tool in the detection and diagnosis of defects in photovoltaic (PV) modules, addressing challenges faced by conventional methods which can be inefficient, especially for large-scale PV arrays. PV modules are susceptible to various defects during manufacturing, transportation, installation, and operation, which can impact their performance and longevity. Deep learning techniques, particularly convolutional neural networks (CNNs), have been increasingly utilized to automate the detection of these defects [1,2].

The application of DL in fault detection of PV systems is crucial for maintaining efficiency and safety, as faults can lead to reduced power output and potential fire risks [3]. Various DL techniques, such as convolutional neural networks (CNNs), have been employed to analyze different types of images—electroluminescence (EL), thermal, and visible spectrum—for identification and classification of defects in PV modules [2–7].

Interestingly, while DL methods have shown high accuracy in defect

detection, challenges persist, such as the need for large datasets and the ability to generalize across different types of PV modules and environmental conditions. For instance, the application of EL images with semantic segmentation models has been explored to detect defects unaided by the equipment applied for image generation [6]. Similarly, CNNs like VGG-16 have been found to perform well in diagnosing defects using thermal images. Moreover, the integration of DL with unmanned aerial vehicles (UAVs) for automated inspection of PV modules has demonstrated efficiency and reliability [5]. However, the fusion of multi-wavelength images did not significantly improve fault classification, indicating an area for further research [8]. Additionally, the presence of external interferences poses a challenge that requires sophisticated image processing techniques to overcome [7].

One approach to defect detection involves the use of electroluminescence (EL) images, which can reveal defects which tend to be less visible to the naked eye. Researchers have developed deep learning-based solutions that combine traditional image processing with generative adversarial networks (GANs) to generate high-quality EL image samples from limited data, followed by the application of convolutional

* Corresponding author.

E-mail addresses: kmasita@uj.ac.za (K. Masita), alinabeal1981@gmail.com (A. Hasan), tshongwe@uj.ac.za (T. Shongwe), hasanabuhilal@hotmail.com (H.A. Hilal).

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neural networks (CNNs) for automatic defect classification [8,9]. This method has shown to improve the performance and effectiveness of PV module assessments and quality management.

Another study has explored the usage of aerial images captured from UAVs for defect pattern recognition in large-scale PV farms. By employing CNNs in a supervised learning process, the system can extract deep features of operating PV modules, enhancing the effectiveness and precision of asset monitoring and quality inspection [10].

Thermal infrared imaging has also been utilized in conjunction with machine learning approaches, such as the VGG-16 and MobileNet models, for detection and classification of defective PV modules. This method has demonstrated a promise of cost reductions in defect classification compared to other existing methods [6].

Furthermore, researchers have proposed novel frameworks for automatic defect detection and classification that combine deep feature extraction through neural networks with classical machine learning techniques. These frameworks have achieved state-of-the-art results in identifying PV cell defects with high accuracy [7]. Semantic segmentation models, such as the U-net architecture, have been trained to analyze EL images for defect detection and quantification, independent of the equipment used to generate the images or the wafer-based module design [11]. Additionally, end-to-end deep learning pipelines have been developed for detection, location, and segmentation of cell-level anomalies in PV modules [12].

In production lines, deep learning-based automatic detection systems have been implemented to identify multitype defects in PV modules, showing promising potential for adoption in real production environments [13]. Edge computing has also been integrated with unmanned aerial vehicles to enable non-destructive and automatic visual inspection systems for large-scale PV plants [14].

Efficient cell segmentation techniques and CNN-based defect detection systems with pseudo-colorization have been designed to automate the process of visual defect inspection, achieving high accuracy in defect detection [15]. Lastly, deep-learning-based methods have been proposed to solve the challenges of scant data and data imbalance, improving the performance of defect detection in PV cells [16].

In summary, deep learning has emerged as a powerful tool for the automatic detection of defects in PV modules, offering solutions that are not only accurate and efficient but also adaptable to various imaging techniques and operational scales.

In summary, DL techniques have proven effective in detecting and classifying defects in PV modules, with various studies demonstrating the potential of CNNs and other DL models in analyzing EL, thermal, and visible images. Despite the progress, there is a need for continued development to overcome challenges such as dataset variability and model generalization. The integration of DL with aerial inspection technologies also presents a promising avenue for automating PV module maintenance. Moreover, the use of high-resolution EL images with convolutional neural networks (CNNs) for defect detection and pseudo-colorization enhances visualization and inspection accuracy [15]. The continuous advancements in this field hold great promise for enhancing the reliability and performance of solar energy systems.

The main contributions of this article are outlined in the below:

- 1) **Comprehensive Review of DL Techniques:** The article provides an extensive review of various deep learning models and techniques, particularly convolutional neural networks (CNNs), used for detecting and classifying defects in PV modules. It discusses the strengths and limitations of models like MobileNet, VGG-16, YOLO.
- 2) **Imaging Techniques for Data Acquisition:** It details various imaging techniques used for data acquisition, including electroluminescence (EL) imaging, infrared thermography, field inspection, drone or aerial imaging, in-line inspection during manufacturing, and time-lapse photography. Each method's advantages and specific applications are outlined.

- 3) **The importance of Preprocessing:** The article emphasizes the importance of preprocessing steps such as image normalization, registration, and segmentation to enhance defect detection accuracy. Techniques for thermal and RGB image processing, including radiometric calibration, noise reduction, and contrast enhancement, are discussed to improve image quality and defect visibility.
- 4) **Advanced Techniques for Model Improvement:** The article highlights advanced techniques like transfer learning, attention mechanisms, and recursive feature elimination (RFE) for improving model accuracy and robustness. It also discusses the use of semantic segmentation models like U-net for defect detection and quantification.

The remainder of this article is organized as follows: Section II covers the literature review, focusing on the background of PV module defect data acquisition, data processing, and defect analysis. Section III discusses various deep learning approaches in defects detection of PV modules, comparing and identifying key takeaways in terms of the strengths and shortcomings of the individual methods. Section IV delves into an analysis of PV module thermographic image data from a 82.5 MW_{DC} Solar Plant in order to demonstrate how thermal delta can be useful in the categorization of different anomalies and their severities. Section V concludes the article.

2. Background on data acquisition, processing and defects detection process flow

2.1. Data acquisition

Collecting image data to detect defects in photovoltaic (PV) modules is vital for maintaining the efficiency and lifespan of solar energy systems. Techniques such as electroluminescence (EL), infrared thermography, and daylight electroluminescence imaging are utilized to identify and analyze defects that can impact PV module performance. The next section reviews various data acquisition methods used in PV module defect detection [8].

2.1.1. Electroluminescence imaging for defect detection

EL imaging is effective for high-resolution inspection of PV modules, enabling the detection of fine defects [17]. Deep learning techniques, specifically Convolutional Neural Networks (CNNs), have been applied to EL images for automatic defect detection with high accuracy [17–19]. Semantic segmentation using deep learning can quantify multiple defects in EL images, improving the reliability of PV modules [20].

2.1.2. Infrared thermography for outdoor defect analysis

Improved outdoor infrared thermography schemes have been developed to modulate PV module temperature, revealing defects more clearly in outdoor conditions [21]. Infrared imaging combined with module optimizers can verify and quantify the impact of defects on PV module performance [22].

2.1.3. Field inspection

Perform visual inspections of PV modules in real-world settings such as solar farms or rooftops. Use high-resolution cameras to capture images of the modules from various angles and distances. This can also include thermal imaging using thermal cameras to capture infrared images of PV modules, which can reveal defects such as hotspots caused by cell cracks or faulty connections.

2.1.4. Drone or aerial imaging

These approaches utilize drones or other aerial imaging platforms equipped with cameras to capture RGB images of PV modules from an overhead perspective. This enables comprehensive inspections of large areas and difficult-to-reach locations. This can also include thermal imaging using thermal cameras to capture infrared images of PV modules, which can reveal defects such as hotspots caused by cell cracks or

faulty connections.

2.1.5. In-line inspection during manufacturing

Incorporate cameras into the manufacturing process of PV modules to capture images of each module as it is being produced. This allows for early detection of defects and quality control measures [6].

2.1.6. Time-lapse photography

Set up cameras to capture time-lapse sequences of PV modules over an extended period. This can help in identifying defects that may develop or worsen over time, such as cracks or delamination.

2.1.7. Manual and automatic annotation

Annotate the acquired images to label the location and type of defects present. This step is essential for training the deep learning model to accurately identify defects.

SUMMARY

By combining the approaches discussed, it is possible to acquire a diverse dataset of image data representing various types of defects commonly found in PV modules. This dataset serves as the foundation for training a deep learning model capable of detecting defects with high accuracy and reliability.

2.2. Data processing

Data preprocessing is a critical step in the analysis of photovoltaic (PV) defects using thermal and RGB images. The papers reviewed provide various methodologies for preprocessing such images to enhance defect detection and evaluation. López-fernández et al. (2017) [23] describe a process involving digital image preprocessing, photogrammetric, and computer vision algorithms to generate a georeferenced 3D point cloud from RGB imagery, which is then combined with thermographic sensor data for a comprehensive 5D analysis. Moshaei-Nezhad et al. (2022) [24] discuss preprocessing to normalize intensity levels between infrared thermography (IRT) and visible light images, followed by similarity and non-rigid transformations for accurate registration. Liu et al. (2011) [25] introduce a technique that combines infrared and visible-light images for segmentation, using the LOG operator to define edges in infrared images and chain code to segment the corresponding visible-light images.

Interestingly, while López-fernández et al. (2017) [23] do not require parameter configuration, thus contributing to the automation of PV farm inspection, Liu et al. (2011) [25] focus on the elimination of noise and the simplification of edge representation in infrared images to improve segmentation efficiency. Moshaei-Nezhad et al. (2022) [24], on the other hand, emphasize the importance of preprocessing for accurate registration in intraoperative imaging, which is crucial for subsequent analysis.

In summary, the reviewed papers highlight the importance of preprocessing in the analysis of PV defects using thermal and RGB images. Preprocessing steps such as image normalization, registration, and segmentation are essential for accurate defect detection and sizing. These steps help in overcoming challenges associated with image analysis, such as noise, intensity variations, and misalignment, thereby enhancing the reliability of the defect detection process.

2.2.1. Thermal images processing

Radiometric Calibration: Conversion of raw thermal data to temperature values using calibration parameters provided by the thermal camera manufacturer.

Noise Reduction: Utilizing filters or algorithms to diminish noise and enhance image quality. Methods like median filtering or Gaussian smoothing are effective for this process.

Normalization: The process of normalizing temperature values to a common scale to remove variations caused by environmental factors like sunlight or camera settings.

Contrast Enhancement: The process of adjusting the image contrast to highlight temperature differences between defect areas and background.

Region of Interest (ROI) Extraction: Identification and extraction of regions containing potential defects for further analysis.

Resizing and Cropping: Standardization of image dimensions for consistency and efficiency in processing. This includes cropping out irrelevant parts of the image.

Data Augmentation: Generation of additional training data by applying transformations like rotation, flipping, or shifting. This helps improve the robustness of machine learning models.

Image Fusion: Fusion of thermal and RGB images to create a composite image that combines thermal temperature information with RGB colour information for a more comprehensive analysis.

2.2.2. RGB images processing

Color Correction: Adjustment of the white balance and correct colour distortions caused by lighting conditions or camera settings.

Noise Reduction: Application of filters like median filtering or Gaussian smoothing to reduce noise and improve image quality.

Contrast Enhancement: Enhancement of contrast to make defects more visible against the background.

Normalization: Normalization of pixel values to a common scale to remove variations caused by lighting conditions.

Image Registration: Alignment of images if necessary, especially if multiple images need to be compared or fused.

Resizing and Cropping: Standardization of image dimensions for consistency and efficiency in processing. Crop out irrelevant parts of the image.

Data Augmentation: Generation of additional training data by applying transformations like rotation, flipping, or shifting. This helps improve the robustness of machine learning models.

Image Fusion: Fusion of thermal and RGB images to create a composite image that combines thermal temperature information with RGB colour information for a more comprehensive analysis.

2.3. Feature extraction and selection

In deep learning for PV module defects detection, feature extraction and selection play crucial roles in enhancing the model's performance and efficiency. Here are some common methods used in this domain:

2.3.1. Convolutional neural networks (CNNs)

Convolutional Neural Networks (CNNs) form the foundation of numerous image processing tasks, including identifying defects in PV modules. CNNs autonomously extract hierarchical features from raw data, eliminating the need for manual feature engineering. Transfer learning, which involves fine-tuning a pre-trained CNN model for PV module defect detection, is frequently employed to utilize knowledge from extensive datasets such as ImageNet [26].

2.3.2. Data augmentation

Data augmentation techniques such as rotation, flipping, scaling, and cropping are employed to artificially increase the diversity of the training dataset. By generating variations of the original images, data augmentation helps the model generalize better to unseen data and improves its robustness to different types of defects [22].

2.3.3. Feature maps visualization

Visualization techniques such as activation maximization and gradient-based methods help in understanding what features the CNN has learned. By visualizing feature maps corresponding to different layers of the network, researchers can gain insights into which parts of the input images are most relevant for defect detection.

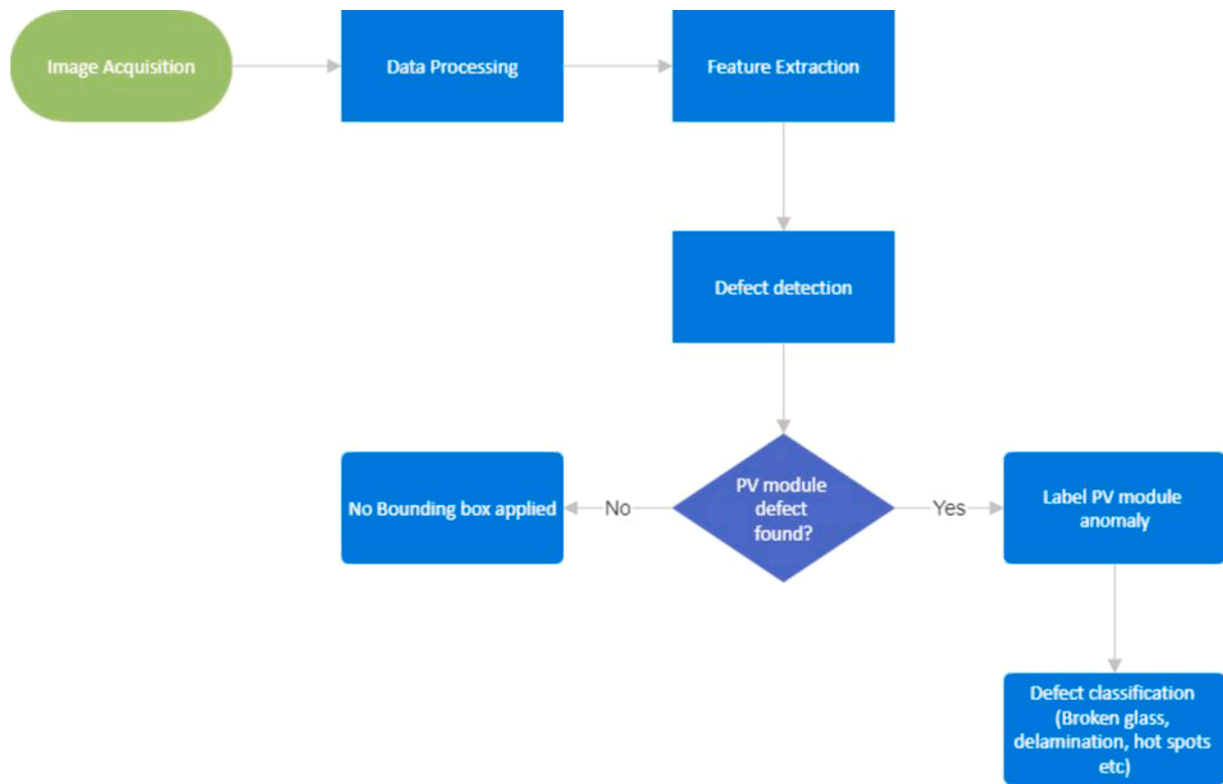


Fig. 1. PV module defects detection process flow.

2.3.4. Autoencoders

Autoencoders are neural networks trained to reconstruct input data. In the context of defect detection, autoencoders can be used for unsupervised feature learning by training them on a large dataset of normal PV module images. The encoder part of the autoencoder can then be used to extract features from PV module images, which can be fed into a classifier for defect detection.

2.3.5. Principal component analysis (PCA)

PCA is a classical method used for dimensionality reduction. While not specific to deep learning, PCA can be applied to reduce the dimensionality of feature vectors extracted from CNNs, especially when dealing with high-dimensional feature spaces. PCA can help in speeding up training and reducing computational costs.

2.3.6. Linear discriminate analysis (LDA)

LDA primarily serves as a feature extraction technique rather than a feature selection method. However, LDA indirectly performs feature selection by transforming the original feature space into a new space with reduced dimensions while maximizing class separability. In the context of defect detection in PV modules, LDA can help identify the most relevant features (e.g., texture, color, shape) that contribute to distinguishing between defective and non-defective modules. By projecting the data onto the discriminant subspace, LDA effectively selects features that maximize the distinction between classes.

2.3.7. Recursive feature elimination (RFE)

RFE is a feature selection technique commonly used in machine learning. It works by recursively removing the least important features until the desired number of features is reached. RFE can be applied to features extracted from CNNs to select the most discriminative features for defect detection.

2.3.8. Attention mechanisms

Attention mechanisms, inspired by human visual attention, help models focus on relevant parts of the input image. By assigning different weights to different spatial locations in the input image, attention mechanisms can effectively highlight regions containing defects. Models with attention mechanisms can be trained end-to-end, allowing the network to learn both feature extraction and selection simultaneously.

SUMMARY

These methods can be used individually or in combination to improve the performance of deep learning models for PV module defect detection. The choice of method depends on factors such as the size and quality of the dataset, computational resources, and the specific requirements of the application.

2.4. Defect detection

The Defects detection stage involves either the use of handcrafted rules or thresholds to identify defects based on extracted features, which is commonly referred to as the Conventional method. Conversely, deep learning or machine learning approaches involve training a machine learning or deep learning model to automatically detect defects using labelled data. It is critical to design, configure or utilize appropriate algorithms such as convolutional neural networks (CNNs) for image-based defect detection.

2.5. Post-processing

Post-processing is essential for refinement of detected defects to remove false positives and improve accuracy. Morphological operations like dilation, erosion, and filtering to refine defect boundaries.

2.6. Defect classification

If defects belong to different classes such as cracks, delamination,

hotspots or shading, machine learning or deep learning models can be used to classify them. This involves the process of training classifiers on labelled defect data to classify detected defects into specific categories.

The flowchart illustrated in Fig. 1 provides a clear and systematic overview of the process involved in automated PV module defect detection and classification, highlighting the key steps and decision points, mainly, data acquisition, data processing and defects detection process as discussed in more detail in this section.

3. Deep learning techniques in PV module defects detection

3.1. Convolutional neural networks (CNNs)

A system that utilizes convolutional neural networks (CNN) was proposed by Zac Brydon et al. for detecting soiling and faults, and it employs digital image processing methods such as HSV conversion, color range extraction, Gaussian filter, and binary threshold for solar panel detection. The system's evaluation showed efficient performance in analyzing single images or smaller collections of images, but it faced challenges in accurately detecting solar panel arrays in less than ideal installation environments and when analyzing videos [27].

In [28] the authors propose a study that emphasizes the importance of performance optimization using transfer learning and explores various training options such as momentum parameter, initial learning rate, data augmentation, mini-batch size, and number of training epochs to achieve the highest classification accuracy. The research outlines the key components of a deep CNN, including the convolutional layer, pooling layer, rectified linear unit (ReLU) layer, and fully connected layers, and highlights the significance of using transfer learning with AlexNet for defect classification. The experimental results indicate the potential of CNN and transfer learning for defect detection in solar panels, with a validation accuracy of about 93.3%.

M. Demirci [29] conducted research on fault detection in photovoltaic (PV) solar panels employing a trained convolutional neural network (CNN). Their CNN-based method categorizes images of PV cells into faulty and normal groups, achieving a 91.1% accuracy rate for binary classification and 88.6% for multi-classification. This study also evaluated the CNN model against other CNN models and pre-trained models, demonstrating its effectiveness in identifying faults in PV systems. The CNN architecture includes convolution layers, pooling layers, fully connected layers, batch normalization, and SoftMax for classification, implemented using MATLAB software.

3.2. MobileNets

Mohamed A. et al. [30] proposed novel solar cells defect detection methodology that utilizes K-means clustering, MobileNetV2, and linear discriminant algorithms to process electroluminescence (EL) images of solar cells. By clustering the cell images and developing a tailored detection model for each cluster, the system can discern between defective and non-defective cells, even with varying shapes. The methodology achieves high accuracy in detecting defects, outperforming recent studies.

Christopher Dunderdale et al. [31] employed the scale invariant feature transform (SIFT) feature descriptor, spatial pyramid matching, and deep learning techniques such as the MobileNets and VGG-16 achieving up to 91.2% accuracy in distinguishing between defective and non-defective PV modules. The study emphasized the necessity of automating the classification process of PV modules, as manual inspection in large-scale plants is impractical. A visual words model, inspired by text sentiment analysis, was utilized to create concise feature sets for machine learning algorithms. The research demonstrated the feasibility of using thermal infrared imagery for the detection and classification of PV module defects, showing potential for increased maintenance efficiency and cost reduction.

3.3. Support vector machines (SVMS)

M. Demirci [32] introduced a Deep Feature-Based (DFB) approach utilizing Support Vector Machines (SVM) for the effective detection of defects in photovoltaic cells through electroluminescence (EL) imaging, achieving high accuracy. The researchers applied the minimum Redundancy Maximum Relevance (mRMR) algorithm for feature selection to minimize feature vector dimensions and enhance classification performance. They identified manual interpretation of EL images as a bottleneck, thus highlighting the need for an automated inspection system. Feature selection is performed by scoring mutual information. The authors show that mutual information between two random variables given as X and Z is calculated as follows:

$$I(X, Z) = \sum_{i,j} P(X=x_i, Z=z_j) \log \frac{P(X=x_i, Z=z_j)}{P(X=x_i)P(Z=z_j)} \quad (1)$$

Where $I(X, Z)$ represents the mutual information factor, while $P(X, Z)$ is the probability distribution between variables X and Z , while $P(X)$ and $P(Z)$ are the probabilistic boundary distribution functions of X and Z , respectively.

Winston et al. [33] conducted a study on detecting anomalies in solar PV panels using advanced methods like Feed Forward Back Propagation Neural Network (FFBPNN) and Support Vector Machine (SVM). The study involved 150 PV modules and aimed to create a fault detection algorithm using ANN and SVM classification tools, focusing on parameters such as open circuit, power loss, and internal imaging. The research classified PV modules into healthy, one hotspot, two hotspots, and micro-cracks, and utilized these classifications to assess the fault detection algorithm's effectiveness. The proposed methodology incorporated the Self-Organizing Map (SOM) and SVM for fault detection, achieving an accuracy rate of 99%. The study compared the proposed fault detection system with an existing fuzzy-based method, highlighting the effectiveness of the SVM approach for accurate fault detection in PV panels. The authors use a Radial Basis Function (RBF) kernel for fault detection and is expressed as:

$$k(X_i, Y_j) = \exp\left(-\frac{\|X_i - Y_j\|^2}{2\sigma^2}\right) \quad (2)$$

Where kernel function 'k', which maps the input space (X_i, Y_j) to the kernel space $\phi(X_i) = \phi(Y_j)$.

$\|X_i - Y_j\|^2$ represents squared Euclidean distance between the two featured vectors, $\frac{1}{2\sigma^2} = \gamma$ gamma Gamma function. Here RBF kernel function is used for the decision region.

3.4. YOLO

Nikhil et al. [34] conducted research using thermal images captured by drones and applied the You Only Look Once (YOLO) method for real-time fault detection and classification. The dataset comprised 277 thermal images, divided into training and validation sets, and labeled with various fault types: temporary hotspot, permanent hotspot, bypass diode fault, and crack/wear and tear. The results demonstrated the method's effectiveness in real-time fault detection and identification, achieving a mean precision of 83.86% [34].

A lightweight YOLO V5 method for detecting thermal spot defects in photovoltaic (PV) panels in infrared images was proposed by Y. Lei et al. [35]. The method involves data enhancement to increase the number of samples in the dataset and improve model generalization, introducing a lightweight network structure (ShuffleNet v2) to replace the YOLO V5 backbone network, and combining the attention mechanism with the neck network to enhance hot spot feature extraction. Experimental results show that the improved network achieves a recognition accuracy of 93.5% for hot spots, which is 4.4% higher compared to the original network. The study compares the proposed method with other models,

demonstrating higher accuracy and confidence in detecting hot spot defects in PV panels.

The study by Tolga Özer and Ömer Türkmen [36] focuses on developing a low-cost AI-based drone for detecting healthy, dusty, and damaged solar panels, utilizing YOLO models (v5, v6, and v8) for classification based on a dataset of 1100 images. Two approaches were explored: Approach 1 involved training models with different YOLO versions, achieving an F1 score of 81% with YOLOv5m, while Approach 2 improved performance using image preprocessing techniques, reaching an F1 score of 87% with YOLOv5s.

3.5. Attention mechanisms

The study in [37] introduces an attention-based deep learning model, Vision Transformer (ViT), for detecting surface defects in renewable energy assets like solar panels and wind turbine blades, achieving over 97% accuracy. ViT leverages the transformer architecture from natural language processing, processing images as sequences of flattened patches to efficiently extract high-level features for defect detection. The model outperforms other deep learning models such as MobileNet, VGG16, Xception, EfficientNetB7, and ResNet50 in terms of accuracy, precision, recall, and execution time.

The research paper in [38] presents a deep learning-based framework for detecting and analyzing cracks in photovoltaic (PV) cells, focusing on microcracks and deep cracks, which affect power output efficiency. The study uses electroluminescence images to train models like U-Net, Attention U-Net, LinkNet, and FPN, with ensemble learning techniques applied to improve segmentation accuracy. The methodology involves classifying cracks based on their orientation and impact on power efficiency, using a dataset from DuraMAT. The models are evaluated using metrics such as mean Intersection-over-Union (mIoU) and F1-Score, with Attention U-Net showing superior performance. Ensemble learning, specifically a weighted average ensemble method, is employed to enhance model performance, resulting in improved mIoU values. This approach assigns weights to models based on their accuracy, giving more importance to better-performing models.

3.6. Autoencoders

Abdellatif et al. [39] proposed faults detection and diagnosis technique for photovoltaic (PV) systems, using a combination of simulation, experimental data, and artificial intelligence (AI) methods. The technique involves the use of an autoencoder (AE) neural network for features extraction from electrical data of PV systems, with the aim of accurately detecting and diagnosing faults in the system. The document outlines the process of database elaboration, including the generation of samples under both faulty and non-faulty conditions, and the use of unsupervised learning methods for features extraction. It also presents the results of the features extraction process, showing the learned features under healthy and faulty operating conditions, as well as the classification accuracy and true positive rate under the presence of noiseless attributes.

Edun et al. [39] the use of spread spectrum time domain reflectometry (SSTDR) and variational autoencoders (VAEs) for detecting and locating faults in solar panel systems. The VAE is trained on non-faulty baseline data to generate an optimal baseline signal without faults, which is then used to compare with the current measurement to detect and localize anomalous fault signatures. The document presents experimental results showing that the VAE approach achieves better results in fault localization compared to using a fixed baseline or the closest baseline choice.

Westraadt et al. [40] developed a deep learning approach using Variational Autoencoders (VAEs) to augment limited datasets for training CNN models, enhancing prediction accuracy. The study utilized thermal images from three different PV plants in South Africa, creating a dataset of 376 thermal images showing modules with hotspot faults and

400 images of non-faulty modules. The classified faults include block faults, patchwork faults, single-cell faults, string faults, and swing faults. The study also proposed combining string and patchwork fault data into a single 'patchwork' fault category.

3.7. Hybrid models

Feng et al. [41] developed a novel framework for detecting defects in photovoltaic (PV) modules by combining visible and infrared images, utilizing deep learning algorithms YOLOv5 and ResNet to enhance detection speed and accuracy. The framework addresses challenges in detecting micro faults in large PV plants, which are complicated by environmental factors and limitations of current infrared camera technologies, achieving a detection speed of 36 frames per second and an accuracy of 95%. Image acquisition is performed using dual infrared cameras to capture both visible and infrared images under consistent conditions, followed by image segmentation of visible images and fault detection on infrared images.

The study in [42] introduces a Vision Transformer (ViT) artificial neural network for fault detection and classification in photovoltaic (PV) modules, achieving high accuracy in identifying anomalies through infrared thermography images. The proposed methodology involves three stages: image preprocessing, data augmentation, and application of ViT, which improves upon existing deep learning techniques for PV fault detection. The ViT model outperforms traditional convolutional neural networks (CNNs) by focusing on holistic image analysis through an attention mechanism, achieving 98.23% accuracy for binary classification and 96.19% for identifying eleven specific PV faults. The research emphasizes the importance of early fault detection in PV systems to maintain performance and prevent risks, leveraging AI-driven fault detection and diagnosis (FDD) models [43].

Jinlai et al. [44]'s study proposes an enhanced YOLOv7-based framework for PV cell anomaly detection, incorporating Partial Convolution and Switchable Atrous Convolution to improve detection accuracy and adaptability to varying defect sizes and complex backgrounds. The PSA-YOLOv7 model outperforms traditional methods and existing CNN-based approaches in terms of detection accuracy, speed, and robustness, making it suitable for real-time applications in solar energy systems. Traditional methods like statistical analysis and signal processing are limited by their need for extensive domain expertise and parameter tuning, whereas deep learning techniques, particularly CNNs, offer superior pattern recognition capabilities. The proposed framework maintains high computational efficiency and real-time detection capabilities, making it applicable for large-scale PV cell monitoring and quality control in solar energy systems. The overall objective of YOLOv7 is to minimize the following loss function:

$$\mathcal{L} = \sum_{i=1}^S \sum_{j=1}^B \left[\lambda_{obj}^{ij} (L_{coord}^{ij} + L_{conf}^{ij}) \right] + \sum_{c=1}^C \lambda_{class}^{ij,c} L_{class}^{ij,c} \quad (3)$$

where S is the number of grid cells, B is the number of bounding boxes per cell, C is the number of classes, λ_{obj} , λ_{class} are the weights for the corresponding terms, $\mathcal{L}_{coord}$, $\mathcal{L}_{conf}$, and $\mathcal{L}_{class}$ are the loss terms for bounding box regression, objectness, and class probability, respectively.

The study in [37] introduces an improved deep learning model, YOLOv8-GD, for detecting defects in electroluminescence images of solar photovoltaic modules, enhancing the YOLOv8 framework with DW-Conv, GSConv, and BiFPN to improve feature extraction and detection efficiency. YOLOv8-GD demonstrates significant improvements in mean average precision and model volume reduction, showing potential for effective photovoltaic defect detection compared to traditional methods and existing models like YOLOv5 and YOLOv8.

Adel Oulefki et al. [45] presented a novel methodology for detecting and analyzing deteriorated areas in solar photovoltaic (PV) modules using unsupervised sensing algorithms and 3D augmented reality (AR),

Table 1
Deep learning based defects in PV modules.

Defect	Defect features	Causes	DL Detection techniques	Percentage Production Loss
Top glass breakage	Visible cracks, shattered glass	Mechanical stress, impact	CNNs, Image segmentation	5–10%
Complete module failure	No power output	Severe damage, electrical failure	CNNs, Anomaly detection	100%
Burn marks	Dark spots, discoloration	Overheating, electrical faults	CNNs, Thermal imaging	10–20%
Encapsulant delamination	Blistering, bubble formation	Aging, poor manufacturing	CNNs, Image classification	5–15%
Cracking	Micro-cracks, line patterns	Mechanical stress	CNNs, Image segmentation	5–15%
Soiling	Dust, dirt accumulation	Environmental factors	CNNs, Image enhancement	10–25%
Shading	Uneven power generation	External obstructions	CNNs, Image classification	20–30%
Hotspot	Localized heating, thermal anomalies	Electrical imbalances	CNNs, Thermal imaging	5–10%
Potential induced degradation (PID)	Power loss, performance degradation	High voltage stress	CNNs, Electroluminescence imaging	10–30%
Snail Trails	Dark streaks, discoloration	Chemical reactions	CNNs, Image classification	5–10%
Overheating junction box	Elevated temperatures, burn marks	Poor connections, electrical faults	CNNs, Thermal imaging	10–15%
Discoloration	Color changes, efficiency reduction	UV exposure, aging	CNNs, Image classification	5–10%

Table 2
Functionality overview of deep learning techniques in PV module defects detection.

DL Method	Functionality	Strengths	Limitations	References
Convolutional Neural Networks (CNNs)	CNNs automatically learn spatial hierarchies of features from input images, making them effective for image classification and segmentation tasks. They use convolutional layers to capture local patterns, pooling layers to reduce dimensions, and fully connected layers for classification. In PV modules, CNNs process EL and thermal images to detect defects like cracks and hotspots.	High Accuracy and Automation: CNNs achieve high accuracy in classifying and segmenting defects such as cracks and hotspots. For instance, Sohail et al. [38] demonstrated a CNN-based method achieving a 91.1% accuracy rate for binary classification of PV cell images.	Data Requirements: CNNs require large datasets for training to achieve high accuracy, which can be a limitation when such datasets are not available. This challenge is noted in the review by Sohail et al. [38].	[38,48]
MobileNet	MobileNet is a lightweight CNN architecture optimized for mobile and embedded applications. It uses depthwise separable convolutions to reduce parameters and computational cost while maintaining accuracy. In PV modules, MobileNet efficiently processes images to identify anomalies with reduced latency and resource usage.	Efficiency and Speed: MobileNet is suitable for mobile and embedded applications, maintaining competitive accuracy while reducing computational cost. Özer and Türkmen (2023) employed MobileNet for solar panel detection using drones.	Reduced Accuracy on Complex Datasets: MobileNet may not perform as well as larger models on complex datasets, potentially limiting its effectiveness in detecting subtle or complex defects.	[49,44]
Support Vector Machines (SVMs)	SVMs classify data by finding the optimal hyperplane that separates different classes. In PV module defect detection, SVMs classify features extracted from images into defective or non-defective categories, often combined with feature extraction techniques to enhance performance.	Robustness in High-Dimensional Spaces: SVMs are effective in high-dimensional spaces and are robust to outliers, making them suitable for classifying features extracted from PV module images.	Feature Dependency: SVMs heavily rely on feature extraction techniques, which can limit their flexibility compared to neural networks. This dependency can be a bottleneck in automating defect detection processes.	[48]
YOLO (You Only Look Once)	YOLO is a real-time object detection system that divides an image into a grid and predicts bounding boxes and class probabilities for each grid cell. It processes the entire image in one pass, making it fast and efficient for detecting multiple defects simultaneously in PV modules.	Real-Time Detection: YOLO is highly efficient for detecting multiple defects simultaneously in PV modules. Cao et al. [37] demonstrated the improved YOLOv8-GD model for defect detection in electroluminescence images.	Challenges with Small Objects: YOLO may struggle with detecting small objects in crowded scenes, which can be a limitation in scenarios where defects are subtle or densely packed.	[37]
Attention Mechanisms	Attention mechanisms enhance neural networks by allowing them to focus on specific parts of the input, improving performance in tasks like image recognition. In defect detection, attention mechanisms help prioritize important features, enabling the model to concentrate on areas of the image that are more likely to contain defects.	Enhanced Feature Extraction: Attention mechanisms allow models to focus on specific parts of the input, improving performance in tasks like image recognition. Dwivedi et al. [37] used attention-based models for identifying surface defects.	Increased Complexity: The use of attention mechanisms can increase computational complexity and requires careful tuning to avoid overfitting.	[37]
Autoencoders	Autoencoders encode input data into a compressed representation and then reconstruct it. They are used in defect detection to highlight anomalies by learning a compact representation of normal data and identifying deviations from this norm. Autoencoders are effective in detecting subtle defects in PV modules.	Anomaly Detection: Autoencoders are effective for unsupervised learning and can highlight anomalies by learning a compact representation of normal data. Westraadt et al. [40] discussed their use for detecting subtle defects in PV modules.	Limited Interpretability: Autoencoders may produce less accurate reconstructions for complex data, and their results can be challenging to interpret without additional context.	[40]

Table 3

Key takeaways on DL techniques and performance.

Technique	Baseline model architecture	Strength	Shortcoming
Automated Framework for Drone-based Solar Panel Soiling Detection [27]	Combination of Convolutional Neural Networks, HSV conversion, Gaussian filtering, and contour creation, to detect arrays, panels, and cells within solar panels for soiling and faults detection.	Machine Learning Integration: The system employs machine learning algorithms to analyze captured images, enhancing the accuracy of detecting soiling or faults in solar panels. Differentiation Capability: It can distinguish between different panels in an array and locate those affected by soiling, which is crucial for targeted maintenance and cleaning. Colour Range Adjustments: Adjustments to color thresholds improve detection rates and help differentiate between various elements of the solar panel arrays.	Fixed Camera Position: The drone camera is locked in a fixed horizontal position, limiting the system's ability to accurately locate soiled panels, especially if the drone's position does not align with the panel's orientation. Wi-Fi Signal Limitations: The video is streamed via Wi-Fi, which constrains the drone's operational range and can be affected by obstacles that interfere with the signal quality. Decreased Accuracy with Increased Soiling: The system's accuracy in detecting cells decreases as the number of cells covered by simulated soiling increases, particularly when soiling patterns split the solar array into multiple sections.
Detection of PV Solar Panel Surface Defects using Transfer Learning of the Deep Convolutional Neural Networks [28]	Feedforward deep CNN with 25 layers, designed for image classification tasks.	High Classification Accuracy: By fine-tuning various training options such as the momentum parameter, initial learning rate, and mini-batch size, the system achieved a high validation accuracy of approximately 93.3%. This indicates strong performance in classifying images of solar panels as defective or normal. Preprocessing and Data Augmentation: The system includes an image preprocessing stage to crop and resize images to the required input dimensions for the CNN. Although data augmentation techniques like reflection, scaling, and rotation did not improve performance in this case, they are generally useful for increasing dataset diversity and reducing overfitting.	Image Quality and Size Constraints: The images collected from online resources may vary in quality and size, which necessitates preprocessing steps such as cropping and resizing. These steps could potentially lead to the loss of important defect features or introduce artifacts that affect the model's performance. Manual Preprocessing Requirement: Each image in the dataset required manual cropping to enclose only the panel surface. This process is time-consuming and may introduce human error or bias in the selection of the region of interest. Data Augmentation Inefficiency: The application of data augmentation techniques such as reflection, scaling, rotation, and flipping did not improve the system's performance and even reduced the validation accuracy by 2%. This suggests that the augmentation strategies may not have been optimally applied or that the model may require more sophisticated augmentation techniques.
A Fault Detection Scheme Utilizing Convolutional Neural Network for PV Solar Panels with High Accuracy [29]	Convolutional neural network model	High Performance Accuracy: The proposed CNN model achieves a high success rate of 91.2% for binary classification (normal vs. faulty) and 88.6% for multi-classification (normal, cracked, dusty, and shadowed). This indicates strong performance in identifying various types of defects in PV panels. Simplicity and Effectiveness: The CNN-based fault detection model is described as straightforward, simple, and effective, suggesting ease of implementation and operation.	Diminished Accuracy with Fewer Layers: The study reports that reducing the number of layers in the CNN model significantly decreases prediction accuracy. For binary classification, accuracy drops to nearly 80%, and for multi-classification, it falls to around 55%, which is substantially lower than the three-layer CNN model's performance. Need for Further Epochs: The study suggests that higher accuracies could be achieved if the simulation were allowed to run for more epochs, indicating that the current model may require additional training time to reach its full potential.
An efficient and portable solar cell defect detection system [30]	Convolutional neural network (CNN) based on the MobileNetV2 framework	Advanced Algorithms: The system employs a combination of K-means clustering, MobileNetV2, and linear discriminant algorithms, which allows for effective feature extraction and classification of solar cell images. Overcoming Shape Confusion: The system can distinguish between defective and non-defective solar cells while overcoming confusion caused by different cell shapes, which is a common issue in solar cell defect detection. Real-Time Detection: The system is capable of real-time defect detection, which is crucial for the ongoing monitoring and maintenance of solar panels to ensure their efficiency. Robust Training Procedure: The model incorporates batch normalization layers	Limited Scope of Defect Detection: The system is designed to detect defects in solar cells using electroluminescence (EL) imaging. It may not be as effective in detecting other types of defects that do not manifest in EL images or require different imaging techniques. Dependence on Image Quality: The accuracy of the system is likely dependent on the quality of the EL images. Poor image quality due to environmental factors or equipment limitations could potentially reduce the system's effectiveness. Data Scarcity and Overfitting: The document mentions the use of data augmentation to overcome data scarcity and overfitting problems. This implies that the system may require a substantial amount of data to train

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Table 3 (continued)

Technique	Baseline model architecture	Strength	Shortcoming
Photovoltaic defect classification through thermal infrared imaging using a machine learning approach [31]	Two deep learning convolutional neural network (CNN) architectures: VGG-16 and MobileNet.	and a fully connected layer with a SoftMax function, contributing to a robust training procedure and accurate classification. High Accuracy: The feature-based classifiers, which include the Scale Invariant Feature Transform (SIFT) descriptor and random forest classifiers, can discriminate between defective and non-defective PV modules with up to 91.2% accuracy. Deep learning models, such as VGG-16 and MobileNet, also show high classification accuracy, with MobileNet achieving up to 89.5% accuracy. Cross-Validation: The system uses a 5-fold cross-validation methodology to ensure that the reported accuracies are indicative of the model's true performance on unseen data, which adds to the reliability of the results.	effectively, and without sufficient data, its performance could be compromised. Training Time: Even with high-end hardware, the training time for convolutional neural networks (CNNs) is significant. This could be a barrier for rapid deployment or iterative development cycles. High Within-Class Variation: Certain defect classes, such as soiling, were not used in the analysis due to high within-class variation and a lack of samples, indicating potential challenges in classifying a wide range of defect types. Erratic Classification Accuracies: The standard deviation in model accuracies suggests that the performance of the models may be inconsistent, which could affect reliability in practical applications.
Efficient deep feature extraction and classification for identifying defective photovoltaic module cells in Electroluminescence images [32]	CNNs along with a proposed L-CNN model tailored	Efficient Feature Extraction: The system uses pre-trained Deep Neural Networks (DNNs) such as DarkNet-19, ResNet-50, VGG-16, and VGG-19 to extract features from the dataset. These features are then combined and optimized using the minimum Redundancy Maximum Relevance (mRMR) algorithm, enhancing the efficiency and effectiveness of feature extraction. Lightweight Architecture: A novel end-to-end Lightweight Convolutional Neural Network (L-CNN) is proposed, which allows for fast learning and efficient training. This makes the system suitable for practical applications where computational resources may be limited.	Class Imbalance: The system must address the challenge of imbalanced class distribution in the EL image dataset, which can affect the performance of machine learning models. Limited Defect Types: The system may need further development to identify a wider range of defect types beyond those mentioned, such as PID, soldering malfunctions, and lamination flaws. Need for Further Optimization: The proposed Lightweight CNN (L-CNN) is trained from scratch without any initial weights, which may require further optimization to achieve the best possible performance.
Solar PV's Micro Crack and Hotspots Detection Technique Using NN and SVM [33]	Three-layered Feed Forward Back Propagation Neural Network (FFBPNN)	High Accuracy: High Accuracy: The system demonstrates high classification accuracy in detecting faults in PV panels. The Support Vector Machine (SVM) method achieves an average accuracy of 99%, which is significantly higher than the Feed Forward Back Propagation Neural Network (FFBPNN) method, which has an average accuracy of 87%. Detection of Multiple Fault Types: Unlike some previous studies that focus only on hotspot detection, this system is capable of identifying both hotspots and micro-cracks, which are small fissures that can negatively impact the lifetime and performance of solar PV systems. Binary Tree Architecture for SVM: The multi-class SVM is implemented using a binary tree architecture, which simplifies the classification process for multiple output classes and contributes to the high accuracy of the system.	Manual Creation of Faults for Testing: Some of the faults (hotspots and micro-cracks) were manually created for the study. This may not fully represent the complexity and variability of faults that occur naturally in the field. Limited to Specific Fault Types: The system is designed to detect hotspots and micro-cracks. Other types of faults that may affect PV panels, such as delamination, snail trails, or Potential Induced Degradation (PID), are not addressed by this system.
Detection and Identification of faults in a PV Module using CNN based Algorithm [34]	Convolutional Neural Network (CNN) based on the "You Only Look Once" (YOLO) algorithm	Good Handling of Different Fault Sizes: The system is capable of detecting faults of varying sizes. It can easily distinguish smaller faults like cracks and also detect larger faults such as temporary hotspots with high accuracy.	Need for Diverse Image Angles: The precision of the model can be improved by including more images taken from different angles. This suggests that the current dataset may not be sufficiently varied in terms of perspectives, which can affect the model's ability to generalize to new, unseen images. Environmental Factors: The system's performance may be influenced by environmental factors that are not accounted for in the thermal images. These unaccounted factors could potentially affect the accuracy of fault detection.

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Table 3 (continued)

Technique	Baseline model architecture	Strength	Shortcoming
A lightweight YOLO V5 method for detecting thermal spot defects in PV panels in infrared images [35]	YOLO V5 object detection algorithm.	Efficient Detection Speed: The system improves network training speed and reduces the model's parameter count by introducing the ShuffleNet v2 lightweight classification network as the backbone. This leads to faster detection speeds. Robust Feature Extraction: The integration of the Coordinate Attention Block Module (CBAM) attention mechanism into the YOLO V5 neck network enhances the model's ability to extract features of small target hot spots, which is crucial for accurate detection. Enhanced Accuracy: The improved network achieves a recognition accuracy of 93.5% for hot spots, which is a 4.4% increase compared to the original YOLO V5 network.	Limited to Infrared Images: The proposed method is specialized for infrared images. If defects are not visible in the infrared spectrum or if infrared imaging is not available, the system may not be applicable. Potential for Missed Detections: Although the system achieves a high recall rate of 95.9%, there is still a possibility of missed detections. This could be critical in scenarios where detecting every single hot spot is essential for the maintenance and operation of PV panels. Generalization to Different Conditions: While data augmentation is used to improve the model's generalization, the system's performance in varying environmental conditions, such as different weather or lighting conditions, is not explicitly discussed.
Deep learning method based on autoencoder neural network applied to faults detection and diagnosis of photovoltaic system. Simulation Modelling Practice and Theory [50]	Deep Neural Network (DNN) that integrates Autoencoders (AEs) with an Artificial Neural Network (ANN) for the purpose of fault detection and diagnosis in photovoltaic (PV) systems.	Robustness to Environmental Variability: By incorporating daily solar irradiance profiles and temperature data, the system can simulate the behavior of PV systems under different environmental conditions. This makes the fault detection process more reliable and adaptable to varying environmental factors. Effective Feature Extraction: The use of unsupervised learning through Autoencoders (AEs) allows for the extraction of new representations from raw inputs. These representations are then used for fault classification, which is crucial for reducing the dimensionality of the data and improving the performance of the fault detection system. Comprehensive Testing: The system has been tested and compared with classical Artificial Neural Networks (ANN) for both noiseless and noisy attributes, demonstrating its effectiveness in various conditions.	Overfitting with Large Databases: The Probabilistic Neural Network (PNN) method, while accurate, has shown drawbacks, particularly with large databases, leading to overfitting. Overfitting can result in a model that performs well on training data but poorly on unseen data. Noise Sensitivity: Although the proposed technique has been tested with noisy attributes, the presence of noise can still lead to misclassifications. The system's performance under noisy conditions is not as robust as with noiseless data, indicating that further improvements could be made to handle noisy environments better.
Anomaly Detection of Disconnects Using SSTDR and Variational Autoencoders [39]	Variational Autoencoder (VAE)	Dynamic Baseline Generation: Unlike traditional methods that use a fixed baseline, the VAE can generate a baseline dynamically in an unsupervised manner. This allows for more accurate fault localization by accounting for changes in the system or environment over time. Robust to Environmental Changes: The system is designed to be robust against environmental effects that typically impact reflectometry responses, such as illuminance, temperature, and humidity.	Non-Linear Fault Effects: Assumptions that faults are a linear combination of some basis signals may not always hold true, especially in the presence of environmental factors like moisture that can change the velocity of propagation (VOP) of the SSTDR signal in a non-linear way. Complexity of Reflectometry Responses: Multiple reflections from the system can make the reflected signal complex to analyze, necessitating advanced algorithms for accurate interpretation.

aiming to enhance solar PV system efficiency by automating anomaly detection. The proposed method outperforms existing segmentation techniques, such as Weka and SAM, in computer simulations, achieving high accuracy in identifying anomalies like hotspots and snail trails, which are often caused by cracks in PV modules. The integration of 3D AR enhances the visualization and localization of faults, providing a more immersive experience for PV system analysis and maintenance, potentially reducing costs and improving repair quality [43,46].

Table 1 below helps to provide a comprehensive overview of various techniques used for detecting faults and defects in photovoltaic (PV) solar panels using machine learning and deep learning models. It details several approaches, including Convolutional Neural Networks (CNNs), transfer learning, and other specialized algorithms, highlighting their strengths and shortcomings [46,47].

The techniques compared vary significantly in their architecture, ranging from deep CNNs with transfer learning capabilities to lightweight models like L-CNN and YOLO V5. Each method demonstrates

high accuracy in specific scenarios; for instance, the YOLO V5 method achieves a recognition accuracy of 93.5% for hot spots, while the Solar PV's Micro Crack and Hotspots Detection Technique using SVM attains an average accuracy of 99%. However, common limitations include the need for large, diverse datasets, potential overfitting, manual pre-processing requirements, and dependencies on specific imaging techniques.

Despite these limitations, it is critical to highlight the potential of these advanced techniques to improve the maintenance and efficiency of solar panels. Techniques like transfer learning and data augmentation are noted for their roles in enhancing model performance and generalization. The tabulated comparison also underscores the importance of robust preprocessing and feature extraction methods, such as the use of Autoencoders (AEs) and the mRMR algorithm, to improve fault detection accuracy. Overall, these approaches represent promising advancements in the field of PV solar panel defect detection, with ongoing research needed to address existing challenges and optimize

Table 4

Deep learning based defects detection techniques.

Ref	Method	Data Source	Input	Detected defects	Average Precision	Contributions
[49]	YOLOv4 and YOLO-CNN-29	FLIR Duo Pro R thermal camera embedded with NVIDIA Jetson Nano and 3-axis gimbal	Thermal and RGB images	Solar Cell failures	100%	The study explores the integration of 5G technology with AI to enhance the operation of drones for long-range inspections. This allows for real-time transmission of the location of defected cells to a ground station. Introduction of a deep learning-based fine-tuned YOLOv4 instance for defect detection on images of extracted modules.
[51]	YOLOv4	Two large-scale PV sites using two UAV models: DJI Phantom 4 and DJI M200	UAV RGB imagery	Glass breakages; Discolorations; and Birds' droppings	83%	
[52]	Deep residual neural network (DRNN)	Manual images of PV panels	RGB images	Dust particles and dust accumulation.	78.7%	Improved and enhanced identification method for uneven dust accumulation.
[53]	Deep CNN	Infrared Solar Modules Dataset	IR images	Diode and connection loss; Cell or multiple cells; Soiling; Cracking; Hotspot and multiple Hotspots; Partial shading; Offline module	92.5%	Reduced operation and maintenance costs.
[54]	YOLOv3-tiny model	UAV IR images	IR images	Hotspot faults	96.5%	Utilization of unmanned aerial vehicles for thermal imaging.
[55]	Deep CNN	Production line RGB images	RGB images	Solar cell busbars; Solar cell fingers; Hotspots	90%	Automatic detection of faults in PV cells.
[56]	YOLOv5+RestNet	UAV images of a PV power station in Hainan Province, China	RGB and IR images	Hotspots; shadowing; Bird droppings	92.1%	Implements photovoltaic (PV) defect detection by fusing visible and infrared images. This approach leverages the strengths of both types of images to enhance the detection of defects in PV modules.
[57]	RGR-Net	MHHS dataset	IR images	Hotspots	89.9%	Cross-embedding Refined Graph Reasoning (CRGR) Module: A key component of the RGR-Net is the CRGR module, which enhances feature representation by integrating spatial and semantic information and establishing global long-range dependencies.
[58]	CNN + SVM		RGB and IR images	Dirt vegetation snail hot cell open Circuit cell crack DC box component	97.52%	Employs a method to create a PV plant information pattern by stitching local images to form a panorama, detecting feature points with the Scale Invariant Feature Transform (SIFT), and using homography transformation and random sample consensus (RANSAC) to match features.
[59]	YOLOv4+light CNN	Ground-based	EL images	Cracks, dirt	99.36%	System is capable of completing the inspection process of an EL image of a PV module within 1.5 s, meeting the requirement for in-line inspection in the PV industry.

performance.

4. Defect detection based on thermal analyses with deep learning

Table 2 below provides a summary of drone data captured thermal data related to PV module defects detection in infrared (IR) images of modules installed in an 82.5 MW_{DC} Solar Plant. The data focuses on actual module temperatures (T1), hotspot temperatures (T2), and the thermal delta (ΔT), which is the temperature difference between T1 and T2. The severity of each defect is classified as either "Critical" or "Minor," and the data includes the irradiance levels and anomaly types for each module.

Key findings from the data indicate that critical severity anomalies often show a higher thermal delta, such as a cracked module with ΔT of 4.40 °C and an isolated underperforming module with ΔT of 3.90 °C. In contrast, minor severity anomalies exhibit lower thermal deltas, such as a module affected by vegetation growth with ΔT of 0.60 °C. The irradiance levels vary across the modules, however, higher irradiance often corresponds to more accurate and reliable results in module anomalies.

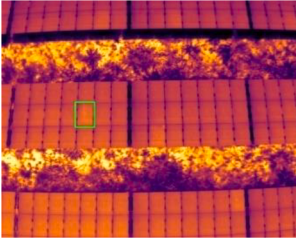
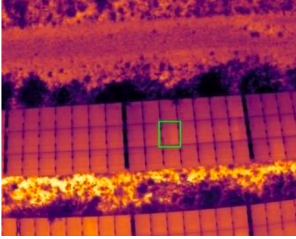
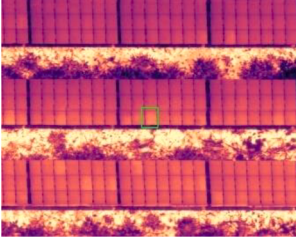
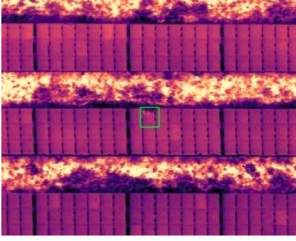
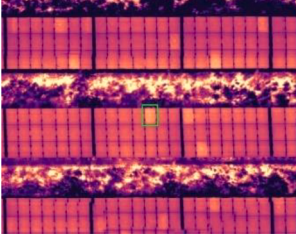
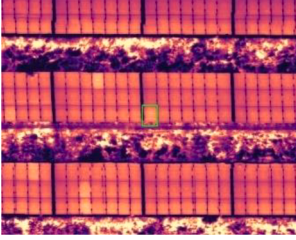
Overall, the Table highlights the importance of monitoring and analyzing thermal deltas and irradiance levels carefully to identify and classify anomalies in module performance. The data reflects a correlation between higher thermal deltas and critical severity, suggesting that addressing these critical defects is essential for maintaining optimal module efficiency and longevity (Tables 3–5).

5. Conclusion

This research work reviews the application of deep learning (DL) techniques in detecting defects in photovoltaic (PV) modules, emphasizing the automation and accuracy improvements over traditional manual inspection methods. Various DL techniques, particularly convolutional neural networks (CNNs), are discussed for their effectiveness in analyzing different types of images, such as electroluminescence (EL), thermal, and visible spectrum, to identify and classify defects. Despite the high accuracy of DL methods, challenges like the need for large datasets and model generalization across different PV modules and environmental conditions persist.

Several imaging techniques for data acquisition, including EL imaging, infrared thermography, field inspection, drone or aerial imaging, in-line inspection during manufacturing, and time-lapse photography are detailed. Each method's advantages and specific applications are outlined. The importance of data preprocessing, such as image normalization, registration, and segmentation, is emphasized to enhance defect detection accuracy. Techniques for thermal and RGB image processing, including radiometric calibration, noise reduction, and contrast enhancement, are discussed to improve image quality and defect visibility. Additionally, the study reviews feature extraction and selection methods, such as CNNs, data augmentation, autoencoders, and principal component analysis (PCA), to improve model performance and efficiency. The accuracy of thermal cameras used in this study is within ± 2 °C for standard infrared thermography, this is due to the high-

Table 5
Anomalies processing based on thermal delta.

Defects detection in IR image	Actual module temp. (T1) [°C]:	Hotspot temp. (T2) [°C]:	Thermal delta $\Delta T = T1-T2$ [°C]:	STC thermal delta $\Delta T = T1-T2$ [°C]:	Severity	Irradiance [W/m ²]:	Anomaly type
	63.40 °C	59.00 °C	4.40 °C	3.89 °C	Critical	1068.00 W/m ²	Cracked modules
	43.50 °C	44.50 °C	1.10 °C	1.52 °C	Minor	806.00 W/m ²	Hotspot affected module
	59.10	63.00 °C	3.90 °C	3.43 °C	Critical	1090.00 W/m ²	Isolated/ Underperforming module
	63.30 °C	63.90 °C	0.60 °C	0.53 °C	Minor	1084.00 W/m ²	Vegetation growth causing hotspots
	51.90 °C	57.30 °C	5.40 °C	5.64 °C	Critical	986.00 W/m ²	Bypass Diode activated module
	56.00 °C	61.10 °C	5.10 °C	5.32 °C	Critical	972.00 W/m ²	Multi Hotspot affected module

precision and calibration of the cameras for the specific environmental conditions involved. The accuracy of the thermal measurement system directly impacts the reliability of defect detection, particularly for subtle temperature variations, thus, providing high accuracy results.

Moreover, this work also explores various DL models and

frameworks used in PV module defect detection. It discusses the implementation and performance of models like MobileNet, VGG-16, YOLO, and others in different studies, highlighting their strengths and limitations. Specific techniques such as transfer learning, attention mechanisms, and recursive feature elimination (RFE) are mentioned for

improving model accuracy and robustness. The study concludes that while DL techniques have proven effective in detecting and classifying PV module defects, ongoing research is needed to address challenges like dataset variability, model generalization, thermal delta dependencies, and the impact of different irradiance levels on PV module performance and anomalies. The integration of DL with aerial inspection technologies and advancements in image processing hold promise for further enhancing the reliability and performance of solar energy systems.

For more comprehensive fault localization and classification in PV systems, future research proposes hybrid models integrating transformer and CNN techniques as well as real-time applications. The model's robustness and accuracy of detection can be further improved by addressing its performance in harsh environments, investigating its capacity to identify novel and uncommon fault kinds, and taking into account the integration of more data sources [46].

Future avenues for study include creating inspection procedures that are specifically tailored to the environment, enhancing methods for localizing defects, and combining several image sources to conduct thorough system evaluations. Defect detection can be extended to other module types, additional defect types can be taken into consideration, data scarcity and class imbalance can be addressed, fully automatic instance segmentation for module extraction, data collection from additional PV plants, and the impact of Ground Sampling Distance (GSD) on defect detection can all be studied. Future studies could focus on correlating delta T thresholds with camera accuracy and the specific conditions of PV modules such as irradiance levels and environmental influences, ensuring robust defect detection across diverse scenarios [60].

CRedit authorship contribution statement

Katleho Masita: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Hasan:** Formal analysis, Conceptualization. **Thokozani Shongwe:** Methodology, Investigation. **Hasan Abu Hilal:** Writing – review & editing, Resources, Project administration.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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